

2 Machines & Models

Efficient Algorithms · Wintersemester 2026–27

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CS566 (Wintersemester 2026–27)
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Learning Outcomes

Unit 2: ***Machines & Models***

1. Understand the difference between empirical *running time* and algorithm *analysis*.
2. Understand *worst/best/average case* models for input data.
3. Know the *RAM machine* model.
4. Know the definitions of *asymptotic notation* (Big-Oh classes and relatives).
5. Understand the reasons to make *asymptotic approximations*.
6. Be able to *analyze* simple *algorithms*.

Outline

2 Machines & Models

2.1 Algorithm analysis

2.2 The Word-RAM Model

2.3 Asymptotics & Big-Oh

2.4 Asymptotic Pitfalls

2.5 Teaser: The Maximum subarray problem

2.6 Teaser: Algorithm Design in Action

What is an algorithm?

An algorithm is a sequence of instructions.

think: recipe



More precisely:

e. g. Python script

1. mechanically executable
   ⇝ no "common sense" needed
2. finite description ≠ finite computation!
3. solves a *problem*, i. e., a class of problem instances

$x + y$, not only $17 + 4$

- input-processing-output abstraction



Typical example: *bubblesort*

⇝ not a specific program
but the underlying idea

What is a data structure?

A data structure is

1. a rule for **encoding data** (in computer memory), plus
2. **algorithms** to work with it (queries, updates, etc.)

typical example: *binary search tree*



2.1 Algorithm analysis

Good algorithms

Our goal: Find good (best?) algorithms and data structures for a task.

Good "usually" means

- fast running *time* (can be complicated in distributed systems)
- moderate memory *space* usage

Algorithm analysis is a way to

- compare different algorithms,
- predict their performance in an application

Running time experiments

Why not simply run and time it?



- results only apply to
  - single *test* machine
  - tested inputs
  - tested implementation
  - ...
  - $\neq$ *universal truths*

- instead: consider and analyze algorithms on an abstract machine
  - $\rightsquigarrow$ provable statements for model
  - $\rightsquigarrow$ testable model hypotheses

survives Pentium 4

$\rightsquigarrow$ Need precise model of machine (costs), input data and algorithms.

Data Models

Algorithm analysis typically uses one of the following simple data models:

- **worst-case performance:**
  consider the *worst* of all inputs as our cost metric
- **best-case performance:**
  consider the *best* of all inputs as our cost metric
- **average-case performance:**
  consider the average/expectation of a *random* input as our cost metric

Usually, we apply the above for *inputs of same size* n.

$\rightsquigarrow$ performance is only a **function of** n.

2.2 The Word-RAM Model

Clicker Question



What is the cost of *adding* two d-digit integers?
(For example, for $d = 5$, what is $45\,235 + 91\,342$?)

- **A** constant time
- **B** logarithmic in d
- **C** proportional to d
- **D** quadratic in d
- **E** no idea what you are talking about



→ *sli.do/effalg*

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Machine models

The machine model decides

- what algorithms are possible
- how they are described (= programming language)
- what an execution *costs*

Goal: Machine models should be
detailed and powerful enough to reflect actual machines,
abstract enough to unify architectures,
simple enough to analyze.

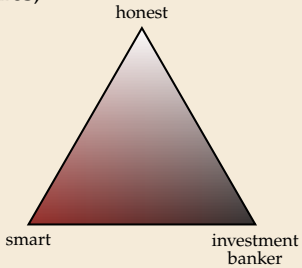
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⇝ usually some compromise is needed

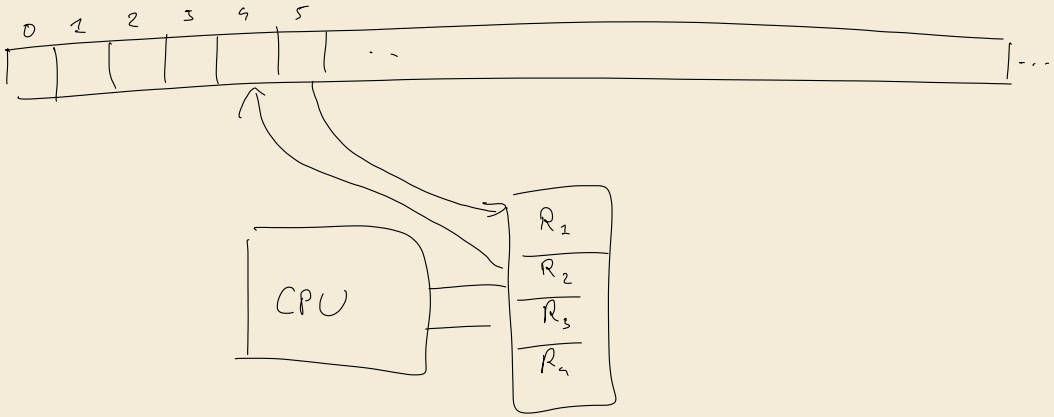


Random Access Machines

Random access machine (RAM)

more detail in §2.2 of *Sequential and Parallel Algorithms and Data Structures* by Sanders, Mehlhorn, Dietzfelbinger, Dementiev

- unlimited *memory* `MEM[0]`, `MEM[1]`, `MEM[2]`, ...
- fixed number of *registers* $R_1, \ldots, R_r$ (say $r = 100$)



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- memory cells $\texttt{MEM}[i]$ and registers R_i store w-bit integers, i. e., numbers in $[0..2^w - 1]$
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  w is the word width/size; typically $\boxed{w \propto \lg n}$ $\rightsquigarrow$ $2^w \approx n$
- Instructions:
  - load & store: $R_i := \texttt{MEM}[R_j]$ $\quad \texttt{MEM}[R_j] := R_i$
  - operations on registers: $R_k := R_i + R_j$ (arithmetic is *modulo* 2^w!)
    also $R_i - R_j$, $R_i \cdot R_j$, $R_i \operatorname{div} R_j$, $R_i \bmod R_j$
    C-style operations (bitwise and/or/xor, left/right shift)
  - conditional and unconditional jumps
- cost: number of executed instructions

we will see further models later

$\rightsquigarrow$ The RAM is the standard model for sequential computation.

RAM-Program Example

Example RAM program

```
1  // Assume: R_1 stores number N
2  // Assume: MEM[0..N) contains list of N numbers
3  R_2 := R_1;
4  R_3 := R_1 − 1;
5  R_5 := R_3 − 1;
6  R_4 := MEM[R_5];
7  R_6 := MEM[R_3];
8  if (R_4 ≤ R_6) goto line 11;
9  MEM[R_5] := R_6;
10 MEM[R_3] := R_4;
11 R_3 := R_3 − 1;
12 if (R_3 > 0) goto line 5;
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14 if (R_2 > 0) goto line 4;
15 // Done:
```

Clicker Question



What algorithm does the RAM program on the previous slide implement?



→ *sli.do/effalg*

RAM-Program Example

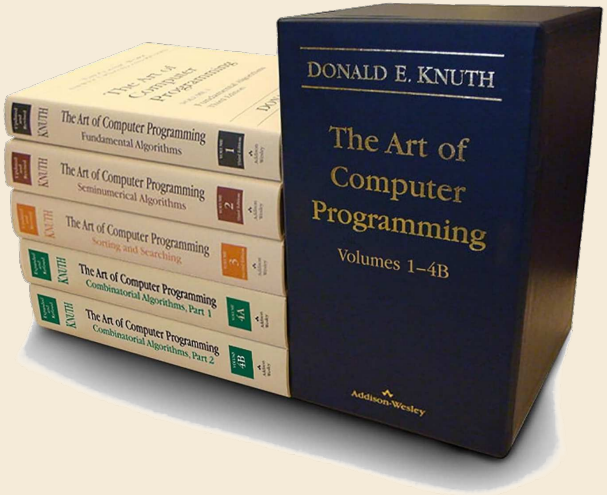
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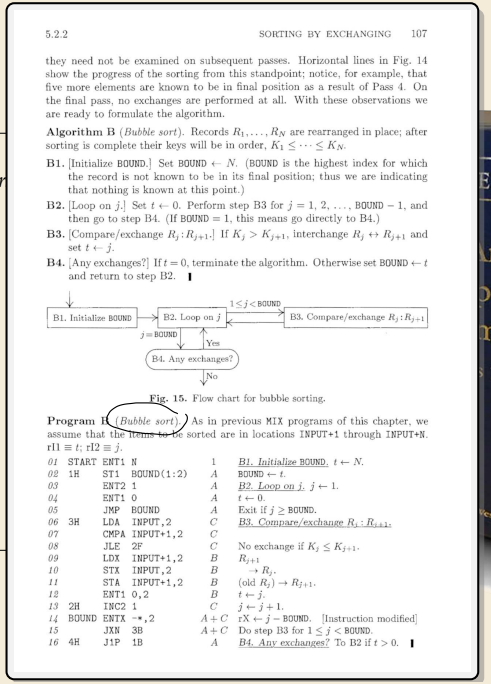
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RAM-Program Example

Example RAM program

1 // *Assume:* R_1 *stores number* N
2 // *Assume:* MEM$[0..N)$ *contains list of* N *numbers*
3 $R_2 := R_1$;
4 $R_3 := R_1 - 1$;
5 $R_5 := R_3 - 1$;
6 $R_4 :=$ MEM$[R_5]$;
7 $R_6 :=$ MEM$[R_3]$;
8 **if** $(R_4 \leq R_6)$ goto line 11;
9 MEM$[R_5] := R_6$;
10 MEM$[R_3] := R_4$;
11 $R_3 := R_3 - 1$;
12 **if** $(R_3 > 0)$ goto line 5;
13 $R_2 := R_2 - 1$;
14 **if** $(R_2 > 0)$ goto line 4;
15 // *Done:* MEM$[0..N)$ *sorted*



5.2.2 SORTING BY EXCHANGING 107

they need not be examined on subsequent passes. Horizontal lines in Fig. 14 show the progress of the sorting from this standpoint; notice, for example, that five more elements are known to be in final position as a result of Pass 4. On the final pass, no exchanges are performed at all. With these observations we are ready to formulate the algorithm.

Algorithm B (*Bubble sort*). Records $R_1, \ldots, R_N$ are rearranged in place; after sorting is complete their keys will be in order, $K_1 \leq \cdots \leq K_N$.

B1. [Initialize BOUND.] Set BOUND $\leftarrow N$. (BOUND is the highest index for which the record is not known to be in its final position; thus we are indicating that nothing is known at this point.)

B2. [Loop on j.] Set $t \leftarrow 0$. Perform step B3 for $j = 1, 2, \ldots,$ BOUND $- 1$, and then go to step B4. (If BOUND $= 1$, this means go directly to B4.)

B3. [Compare/exchange $R_j : R_{j+1}$.] If $K_j > K_{j+1}$, interchange $R_j \leftrightarrow R_{j+1}$ and set $t \leftarrow j$.

B4. [Any exchanges?] If $t = 0$, terminate the algorithm. Otherwise set BOUND $\leftarrow t$ and return to step B2. ∎

Fig. 15. Flow chart for bubble sorting.

Program B (*Bubble sort*). As in previous MIX programs of this chapter, we assume that the items to be sorted are in locations INPUT+1 through INPUT+N. rI1 $\equiv t$; rI2 $\equiv j$.

```
01  START ENT1 N              1     B1. Initialize BOUND. t ← N.
02  1H    ST1  BOUND(1:2)     A     BOUND ← t.
03        ENT2 1              A     B2. Loop on j. j ← 1.
04        ENT1 0              A     t ← 0.
05        JMP  BOUND          A     Exit if j ≥ BOUND.
06  3H    LDA  INPUT,2        C     B3. Compare/exchange R_j : R_{j+1}:
07        CMPA INPUT+1,2      C
08        JLE  2F             C     No exchange if K_j ≤ K_{j+1}.
09        LDX  INPUT+1,2      B     R_{j+1}
10        STX  INPUT,2        B       → R_j.
11        STA  INPUT+1,2      B     (old R_j) → R_{j+1}.
12        ENT1 0,2            B     t ← j.
13  2H    INC2 1              C     j ← j + 1.
14  BOUND ENTX *+*,2          A+C   rX ← j − BOUND.  [Instruction modified]
15        JXN  3B             A+C   Do step B3 for 1 ≤ j < BOUND.
16  4H    J1P  1B             A     B4. Any exchanges? To B2 if t > 0. ∎
```

Pseudocode

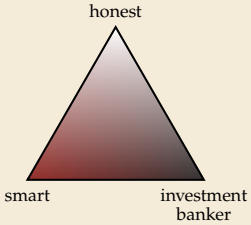
- Programs for the random-access machine are very low level and detailed

≈ assembly/machine language

Typical simplifications when describing and analyzing algorithms:

- more abstract *pseudocode* ← code that humans understand (easily)
  - control flow using **if**, **for**, **while**, etc.
  - variable names instead of fixed registers and memory cells
  - memory management (more below)
- count dominant *elementary operations* (e. g. memory accesses) instead of all RAM instructions

In both cases: We *can* go to full detail where needed/desired.



Pseudocode – Example

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Pseudocode Algorithm

```
1  procedure bubblesort(A[0..N)):
2      for i := N, N − 1, ..., 1
3          for j := N − 2, N − 3, ..., 0
4              if A[j] > A[j + 1]:
5                  Swap A[j] and A[j + 1]
6              end if
7          end for
8      end for
```

- ⇝ much more **readable**
- closer to modern high-level programming languages
- **but:** only allow primitive operations that correspond to $O(1)$ RAM instructions
  - ⇝ analysis

Memory management & Pointers

- A random-access machine is a bit like a bare CPU . . . without any operating system
  - ⇝ cumbersome to use
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    - used to allocate a `new` array (of a fixed size) or
    - a `new` object/record (with a known list of instance variables)
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We will mostly ignore *how* all this works here.

2.3 Asymptotics & Big-Oh

Clicker Question



What is the correct way to complete the equation?

$8n + \frac{1}{2}n^2 + 1024 = \square$

A $O(1)$

B $O(n)$

C $O(n \log(n))$

D $O(n^2)$

E I don't know $O(\cdot)$



→ *sli.do/effalg*

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Why asymptotics?

Algorithm analysis focuses on (the limiting behavior for infinitely) **large** inputs.

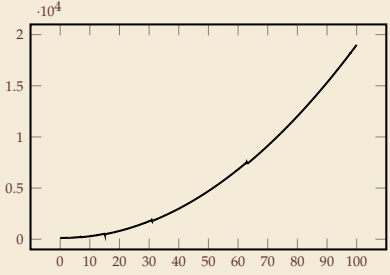
- abstracts from unnecessary detail
- simplifies analysis
- often necessary for sensible comparison



Asymptotics = approximation around ∞

Example: Consider a function $f(n)$ given by

$$2n^2 - 3n\lfloor \log_2(n+1) \rfloor + 7n - 3\lfloor \log_2(n+1) \rfloor + 120$$



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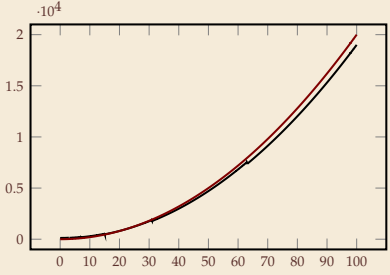
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Asymptotic tools – Formal & definitive definition

- **"Tilde Notation":** $f(n) \sim g(n)$ iff (if, and only if) $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 1$

  „f and g are *asymptotically equivalent*"

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- **"Big-Oh Notation":** $f(n) \in O\big(g(n)\big)$ iff $\left|\frac{f(n)}{g(n)}\right|$ is bounded for $n \geq n_0$

  (also write '=' instead)

  iff $\limsup_{n\to\infty} \left|\frac{f(n)}{g(n)}\right| < \infty$

  (need supremum since limit might not exist!)

Variants:

- $f(n) \in \Omega\big(g(n)\big)$ iff $g(n) \in O\big(f(n)\big)$ ("Big-Omega")
- $f(n) \in \Theta\big(g(n)\big)$ iff $f(n) \in O\big(g(n)\big)$ **and** $f(n) \in \Omega\big(g(n)\big)$ ("Big-Theta")

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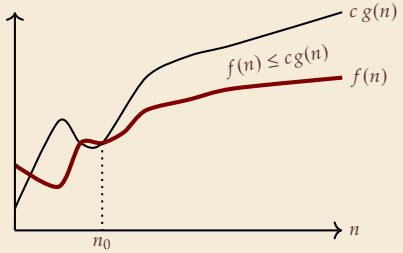
- **"Little-Oh Notation":** $f(n) \in o\big(g(n)\big)$ iff $\lim_{n\to\infty} \left|\frac{f(n)}{g(n)}\right| = 0$

  similarly: $f(n) \in \omega\big(g(n)\big)$ if $\lim = \infty$

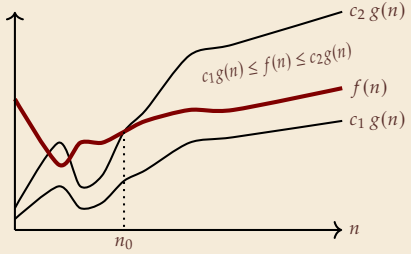
(Benefit of this definition: Works for any $f, g : \mathbb{R} \to \mathbb{R}$ (with $g(n) \neq 0$ for large n) and is easy to generalize to limits other than $n \to \infty$)

Asymptotic tools – Intuition

- $f(n) = O(g(n))$: $f(n)$ is **at most** $g(n)$
  (1) up to constant factors and
  (2) for sufficiently large n



- $f(n) = \Theta(g(n))$: $f(n)$ is **equal to** $g(n)$
  up to constant factors and
  for sufficiently large n



Plots can be misleading!

Example

Clicker Question



Assume $f(n) \in O(g(n))$. What can we say about $g(n)$?

A $g(n) = O(f(n))$

B $g(n) = \Omega(f(n))$

C $g(n) = \Theta(f(n))$

D Nothing (it depends on f and g)



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A ~~$g(n) = O(f(n))$~~

B $g(n) = \Omega(f(n))$ ✓ (if $g(n) \neq 0$)

C ~~$g(n) = \Theta(f(n))$~~

D ~~Nothing (it depends on f and g)~~



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Asymptotics – Example 1

Basic examples: $f(n) = O(g(n))$ $\quad g(n) = n^3$

- $20n^3 + 10n\ln(n) + 5 \sim 20n^3 = \Theta(n^3)$
- $3\lg(n^2) + \lg(\lg(n)) = \Theta(\log n)$
- $10^{100} = O(1)$

$$\frac{f(n)}{g(n)} = \frac{20n^3 + 10n\ln(n) + 5}{n^3}$$

$$= 20 + \underbrace{\frac{10n\ln(n)}{n^3}}_{\to 0} + \underbrace{\frac{5}{n^3}}_{\to 0}$$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 20 < \infty$$

Use *wolfram alpha* to compute/check limits, but also practice it with pen and paper!

Clicker Question



Is $(\sin(n) + 2)n^2 = \Theta(n^2)$?

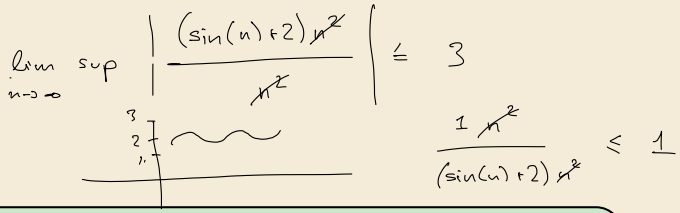
A Yes

B No



→ *sli.do/effalg*

Clicker Question



Is $(\sin(n) + 2)n^2 = \Theta(n^2)$?

A Yes ✓

B ~~No~~

→ *sli.do/effalg*

Asymptotics – Basic facts

Rules to work with Big-Oh classes:

- $f = \Theta(f)$ (reflexivity)
- $f = \Theta(g) \wedge g = \Theta(h) \implies f = \Theta(h)$
- $c \cdot f(n) = \Theta(f(n))$ for constant $c \neq 0$
- $f \sim g \iff f = g \cdot (1 \pm o(1))$
- $\Theta(f) \cdot \Theta(g) = \Theta(f \cdot g)$
- $\Theta(f) + \Theta(g) = \Theta(f + g) = \Theta(\max\{f, g\})$ for $f, g \geq 0$
  largest summand determines Θ-class

$$O(g(n)) = \left\{ f : \limsup_{n \to \infty} \left| \frac{f(n)}{g(n)} \right| < \infty \right\}$$

$$f \in O(g)$$

$$\Theta(n) = O(n^2)$$

$$g(n) \cdot (1 \pm o(1)) =$$

$$\{ h : |h(n) = g(n)(1 + k(n)) \vee h(n) = g(n)(1 - k(n))| \wedge k(n) \in o(1) \}$$

"$f = o(1)$"?

$$o(1) = \left\{ f : \lim_{n \to \infty} \frac{f(n)}{1} = 0 \right\}$$

$$(\max\{f, g\})(n) = \max\{f(n), g(n)\}$$

Asymptotics – Frequently encountered classes

Frequently used orders of growth:

- constant $\Theta(1)$
- logarithmic $\Theta(\log n)$ Note: $a, b > 1$ constants $\rightsquigarrow$ $\Theta(\log_a(n)) = \Theta(\log_b(n))$
- linear $\Theta(n)$
- linearithmic $\Theta(n \log n)$
- quadratic $\Theta(n^2)$
- cubic $\Theta(n^3)$
- polynomial $O(n^c)$ for some constant c
- exponential $\Theta(c^n)$ for some constant $c > 1$; $2^{\Theta(n)}$ Note: $a > b > 0$ constants $\rightsquigarrow$ $b^n = o(a^n)$

2.4 Asymptotic Pitfalls

Asymptotics – Example 2

Square-and-multiply algorithm
for computing x^m with $m \in \mathbb{N}$

Inputs:

- m as binary number (array of bits)
- n = #bits in m
- x a floating-point number

```
1 double pow(double base, boolean[] exponentBits) {
2     double res = 1;
3     for (boolean bit : exponentBits) {
4         res *= res;
5         if (bit) res *= base;
6     }
7     return res;
8 }
```

Iteration order must be most- to least significant bit!

$5 = 101_2$

x^5 ⇝ 1 ⇝ 1 ⇝ x ⇝ x^2 ⇝ x^4 · x

- Cost: C = # multiplications
- C = n (line 4) + #one-bits in binary representation of m (line 5)

⇝ $n \leq C \leq 2n$

$1 \leq \frac{C}{n} \leq 2$

Clicker Question



We showed $n \leq C(n) \leq 2n$; what is the most precise asymptotic approximation for $C(n)$ that we can make?

Write e. g. `O(n^2)` for $O(n^2)$ or `Theta(sqrt(n))` for $\Theta(\sqrt{n})$.



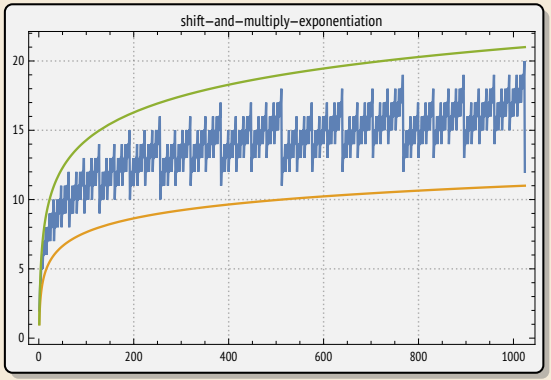
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- $C = n$ (line 4) + #one-bits in binary representation of m (line 5)

$\rightsquigarrow\ n \leq C \leq 2n$

$\rightsquigarrow\ \boxed{C = \Theta(n) = \Theta(\log m)}$



Often, you can pretend Θ is *"like $\sim$ with an unknown constant"*
but in this case, ***no such constant exists!***

Asymptotics with several variables

- **Example:** Algorithms on (connected, simple) graphs with n vertices and m edges.
  - want to say: Algorithm A takes time $\Theta(n + m)$.
  - But what does that even mean, formally?!

Asymptotics with several variables

- **Example:** Algorithms on (connected, simple) graphs with n vertices and m edges.
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  - But what does that even mean, formally?!

⚠ Inconsistent and incompatible definitions used in the literature!

- **Here:**
  - (implicitly) always have a single *"main" variable* n: with $n \to \infty$
  - all other variables are *functions* of n: $m = m(n)$
  - must make *conditions* on functions explicit: $m(n) \in \Omega(n)$ and $m(n) \in O(n^2)$.

  ⇝ Can make statements like

$$O(n+m) \subseteq O(nm) \qquad (n \to \infty,\ m \in \Omega(1))$$

Assume $m(n) = \frac{1}{2^n}$ ↯

2.5 Teaser: The Maximum subarray problem

Bring on the puzzles!

Time for a concrete example of algorithm design!

- we will illustrate the algorithm design process on a "toy problem"
- clean abstract problem, but nontrivial to solve!

Bring on the puzzles!

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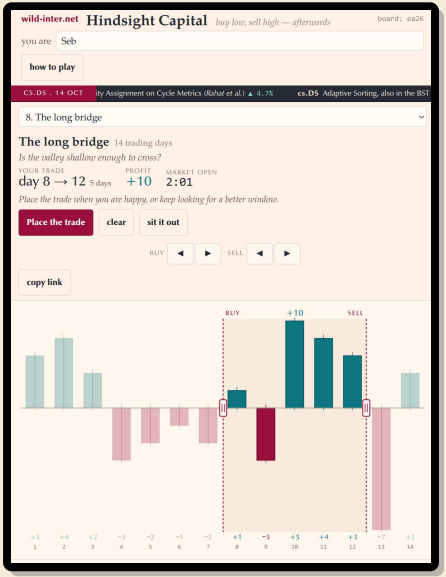
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Maximum (sum) subarray problem

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will ignore that here; easy to modify algorithms

Let’s Play!



tiny.cc/subarray

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Applications:

- largest gain of a stock $A[i]$ price change on day i
- signal detection in biological sequence analysis
- 2D generalization used in image analysis

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will ignore that here; easy to modify algorithms

Modeling decisions:

- input size: # numbers n
- assume all integers (and sums) fit in $O(1)$ words
- ⇝ count # additions as elementary operation

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- largest gain of a stock $A[i]$ price change on day i
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Clicker Question



What do you think is the Θ-class of the running time of the fastest algorithm for the maximal sum subarray problem?



→ *sli.do/effalg*

Template for Describing an Algorithm

(D&C, Greedy)

1. **Algorithmic Idea**
   Abstract idea that makes the algorithm work (prose)
   (an expert could fill in the rest from here)

2. **Pseudocode**
   structured description of procedure including edge cases
   should be unambiguous and close to real code

3. **Correctness proof**
   argument why the correct result is computed
   often uses induction and invariants

4. **Algorithm analysis**
   analysis of the efficiency of the algorithm
   usually want Θ-class of worst-case running time
   where interesting, also space usage

Brute force approach

- Let's start with the simplest thinkable solution

1. 💡 **Algorithmic Idea**
   try all contiguous subarrays $A[i..j)$

Maximal subarray problem

- **Given:** $A[0..n)$ with $A[i] \in \mathbb{Z}$ for $0 \le i < n$.
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10 end for
```

s(i,j)

Maximal subarray problem

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3. Correctness proof

direct by definition of s

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$$= \sum_{i=0}^{n-1} \sum_{j=i}^{n} \sum_{k=i}^{j-1} 1$$

$$= (j-1)-i+1 = j-i$$

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$$= \sum_{i=0}^{n-1}\sum_{j=0}^{n-i} j$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$1+2+3+4+5+6$$

$$+\ 6+5+4+3+2+1$$

$$7\ \ 7\ \ 7\ \ 7\ \ 7\ \ 7 = 7\cdot 6/2$$

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$$= \sum_{i=0}^{n-1} \sum_{j=i}^{n} \sum_{k=i}^{j-1} 1 = \sum_{i=0}^{n-1} \sum_{j=i}^{n} (j-i)$$

$$= \sum_{i=0}^{n-1} \sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1} \frac{(n-i)(n-i+1)}{2}$$

$\frac{1 \cdot 2}{2}$ $\frac{n(n+1)}{2}$

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$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

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$$= \sum_{i=0}^{n-1}\sum_{j=0}^{n-i} j = \sum_{i=0}^{n-1}\frac{(n-i)(n-i+1)}{2}$$

$$= \frac{1}{2}\sum_{i=1}^{n} i(i+1) = \frac{1}{2}\sum_{i=1}^{n} i^2 + \frac{1}{2}\sum_{i=1}^{n} i$$

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$$\begin{aligned}
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&= \sum_{i=0}^{n-1}\sum_{j=0}^{n-i} j \;=\; \sum_{i=0}^{n-1}\frac{(n-i)(n-i+1)}{2} \\
&= \frac{1}{2}\sum_{i=1}^{n} i(i+1) \;=\; \frac{1}{2}\sum_{i=1}^{n} i^2 + \frac{1}{2}\sum_{i=1}^{n} i \\
&= \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4}
\end{aligned}$$

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&= \frac{n(n+1)(n+2)}{6} \;\sim\; \frac{1}{6}n^3 \;=\; \Theta(n^3)
\end{aligned}
$$

Reusing sums

1. **Algorithmic Idea**
   - brute force algorithm is unnecessarily wasteful!
   - can use $s(i, j) = s(i, j-1) + A[j-1]$

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2. **Pseudocode**

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5         t := t + A[j − 1]
6         if t > s then s := t
7     end for
8 end for
```

3. **Correctness proof:** as above

4. **Algorithm analysis:** $\sum_{i=0}^{n-1} \sum_{j=i+1}^{n} 1 = \frac{n(n+1)}{2} \sim \frac{1}{2}n^2 = \Theta(n^2)$ additions

Reusing sums

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Can we possibly do better?

- There are $\binom{n+1}{2} \sim \frac{1}{2}n^2$ different $s(i,j)$ ...
  (one addition each above!)

$\rightsquigarrow$ Can't look at all of them

3. Correctness proof: as above

4. Algorithm analysis: $\displaystyle\sum_{i=0}^{n-1}\sum_{j=i+1}^{n} 1 = \frac{n(n+1)}{2} \sim \frac{1}{2}n^2 = \Theta(n^2)$ additions